



MANIPAL SCHOOL OF INFORMATION SCIENCES

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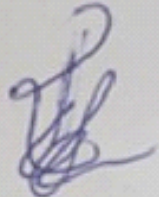
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Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems)

Course Plan

Course Name	:	Advanced Computer Architecture
Course Code	:	ESE 5101
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Ms. Ravikala Kamath
Name of the Program Coordinator	:	Dr. B Dinesh Rao

	
Date:	1.8.2026
Signature of Program Coordinator	
with Date	

	
Date:	30.07.2026
Signature of Course Coordinator	
with Date	



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Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems), program are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in embedded systems.
PEO 2	Introduce state of art technologies in the area of embedded systems and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.



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Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems), graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop and implement embedded systems requirements based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analyzing and decision-making.



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1. Course Plan

1.1 Primary Information

Course Name	:	Advanced Computer Architecture [ESE 5101]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Basic C programming
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
CO1	Apply the knowledge of computer architecture to explore the components of ARM processors.	12	PO3	3
CO2	Analyse the features of ARM processors in designing applications.	15	PO5	4
CO3	Apply the concepts of pipelining and parallel processing for designing efficient applications.	9	PO4	3



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1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	3	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Apply the knowledge	Analyse	Apply the concepts, knowledge
Pattern	Each question carry 3 / 4 / 5 / 10 marks	Assignment question carry 5, 10 marks for CO1, CO2, CO3	Each question carry 3/4/5 marks
Schedule	As per academic calendar.	Assignment submission: 30th aug, 30 sept, 30th oct 2026	As per academic calendar.
Topics covered	Introduction to architecture, component of ARM, ALU design multipliers, pipelines, parallel processors	Register design, ALU, Multipliers, assembly programs, instruction set, control unit, parallel processing hazards	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.



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1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Introduction -Computer architecture Vs organization, classification Von Neman Vs Harvard architecture, SOC, Bus structures , classification RISC vs CISC Vs VLIW	C01
L2	Introduction to ARM Processors: Architecture - Components - Instruction level registers	C01
L3	design of various types of instructions levels	C01
L4	General Purpose Registers – Register Design	C01
L5	ALU– ALU design	C01
L6	Variups types of Adders	C01
L7	Variups types of Adders	C01
L8	Design of Adder circuit	C01
L9	Multiplier – Multiplier Design - Barrel Shifter – Design of Barrel Shifter –	C01
L10	Sequential Multiplication Algorithh	C01
L11	Sequential Multiplication Algorithm	C01



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L12	Sequential Multiplication Algorithm - Booths Algorithm	C01
L13	Division Algorithm (Restoring and Non-Restoring).	C02
L14	Instruction Architecture: ARM & Thumb Instructions , Addressing Modes – Types of Instructions – Endianness	C02
L15	Assembly Programming - example of programs	C02
L16	Assembly Programming - example of programs	C02
L17	ARM assemble programming	C02
L18	Instruction Designing- Huffman Encoding technique for designing instruction sets	C02
L19	Huffman Encoding technique for designing instruction set	C02
L20	Control Unit – control Unit introduction, HARDWARE APPROACH	C02
L21	MSE	C01, C03
L22	deisgn of hardware approach	C02
L23	hardware approach- for example	C02
L24	Microproamming approach	C02
L25	Firmware, Coprocessor, Floating Point Number, FPU	C02



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L26	Types of Memor,Memory Hierarchy, Static, Dynamic RAM, ROM & ROM Types	C02
L27	Cache memories,Performance Considerations, Virtual Memories,	C02
L28	MMU & MPU, Secondary Storage.	C02
L29	Pipelining: Advantag, neccessity of pipelines	C03
L30	Data Hazards, Instruction Hazard	C03
L31	Influence of Instructions sets, Datapath,Control Considerations	C03
L32	Parallel Processing: Parallelism in Uniprocessor	C03
L33	Parallel processing& Multicore Systems	C03
L34	Parallel Computer Structures,	C03
L35	Architectural Classification Schemes, Applications,	C03
L36	Principles and Vector Processing, Structures	C03
L37	Algorithms for Array Processors.	C03



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1.5 References

1. C. V. Hamacher, Z. G. Vranesic, and S. G. Zaky, Computer Organization, 5th ed., Tata McGraw-Hill.
2. M. Rafiquzzaman, Modern Computer Architecture, Chandra, Galgotia Publications.
3. J. L. Hennessy and D. A. Patterson, Computer Architecture: A Quantitative Approach, 2nd ed.
4. J. L. Hennessy and D. A. Patterson, Principles of Computer Architecture, Prentice Hall.
5. S. Danadamudi, Guide to RISC Processors for Programmers and Engineers, Springer Publications.
6. D. Seal, ARM Architecture Reference Manual, 2nd ed., Addison-Wesley.
7. ARM Ltd., AMBA Specification, ARM7TDMI Datasheet.
8. D. A. Patterson and J. L. Hennessy, Computer Organization and Design.
9. D. Seal, ARM Architecture Reference Manual, 2nd ed., Addison-Wesley Professional.
10. S. Furber, ARM System-on-Chip Architecture, 2nd ed., Addison-Wesley Professional, ISBN-13: 978-0201675191, ISBN-10: 0201675196.
11. W. Hohl and C. Hinds, ARM Assembly Language: Fundamentals and Techniques, 2nd ed., ISBN-13: 978-1482229851, ISBN-10: 1482229854.
12. A. Sloss, D. Symes, and C. Wright, ARM System Developer's Guide: Designing and Optimizing System Software, 1st ed., The Morgan Kaufmann Series in Computer Architecture and Design, ISBN-13: 978-1558608740, ISBN-10: 155860874
13. Websites & Transaction Papers.
14. MOOC: <https://www.mooc-list.com/course/computer-architecture-coursera>

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.



1.7 Course Timetable

I Semester						Room: LH01- LH08			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday									
Tuesday									
Wednesday									
Thursday									
Friday									
Saturday									



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1.8 Assessment Plan

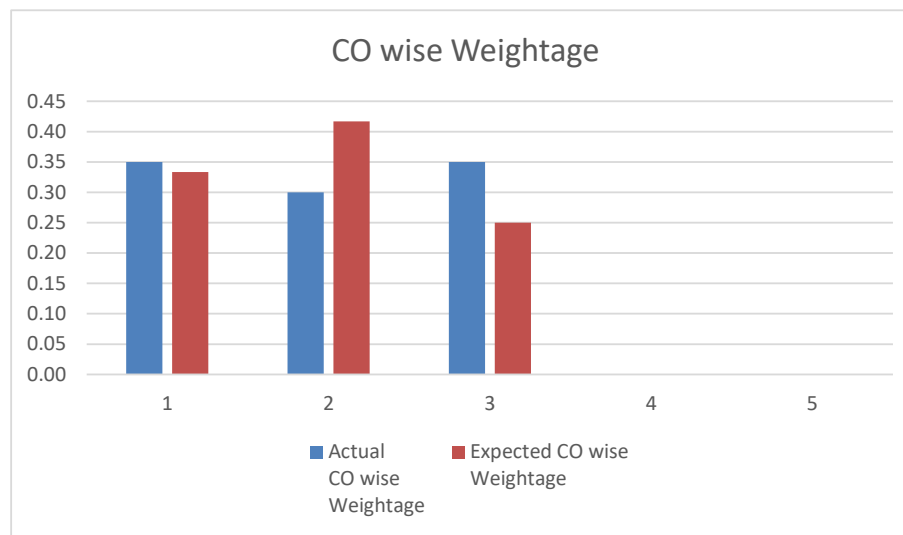
COs		Marks & Weightage				
CO No.	CO Name	MSE Marks (Max. 50)	Assignment Marks (Max. 20)	ESE Marks (Max. 100)	Actual CO wise Weightage	Expected CO wise Weightage
C01	Apply the knowledge of computer architecture to explore the components of ARM processors.	25	5	30	0.35	0.33
C02	Analyse the features of ARM processors in designing applications.	0	10	40	0.30	0.42
C03	Apply the concepts of pipelining and parallel processing for designing efficient applications.	25	5	30	0.35	0.25
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00



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Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes the performances in class participation, assignment work, class tests, MSE, quizzes etc.
- ESE for this course is conducted for a maximum of 100 and the same will be scaled down to 50 for grading.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

$$\text{Weightage for CO1} = (\text{MSE marks for CO1} / 1.6666 + \text{Assignment marks for CO1} / 1.0 + \text{ESE marks for CO1} / 2) / 100$$



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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	- Performance is measured using MSE attainment level. - Reference: question paper and answer scheme. - MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.
2	Assignments	0.2	2	- Performance is measured using assignments/quiz attainment level. - Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	- Performance is measured using ESE attainment level. - Reference: question paper and answer scheme. - ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2					Y
CO3				Y	
Average Articulation Level			*	*	*